

DAY 1-5

Days 1-5: Build and Configure a Small Office Network
Deliverable: Fully functional LAN with: Proper IP addressing and subnetting. Configured DHCP and DNS services. VLANs for network segmentation. Simulated internet access via a router. Troubleshooting documentation for common issues.
Task: Submit a detailed report on the network setup and configurations performed.

1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

Aim

To configure static IP addresses on five devices within the same subnet and verify network connectivity.

IP Address Configuration

Device	IP Address	Subnet Mask	Default Gateway
PC-1	192.168.10.11	255.255.255.0	192.168.10.1
PC-2	192.168.10.12	255.255.255.0	192.168.10.1
PC-3	192.168.10.13	255.255.255.0	192.168.10.1
PC-4	192.168.10.14	255.255.255.0	192.168.10.1
PC-5	192.168.10.15	255.255.255.0	192.168.10.1

Verification

Command Used:

```
ping 192.168.10.12
```

Observation

The ping command returned successful replies with 0% packet loss, confirming that all devices in the same subnet could communicate successfully.

Result

Static IP addresses were assigned successfully, and all five devices communicated correctly within the same subnet.

2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Aim

To configure a DHCP server that automatically assigns IP addresses.

DHCP Configuration

Parameter	Value
Network	192.168.20.0/24
DHCP Pool	192.168.20.100 – 192.168.20.150
Default Gateway	192.168.20.1
DNS Server	8.8.8.8

Assigned Addresses

Device	Assigned IP Address
Laptop-1	192.168.20.100
Laptop-2	192.168.20.101
Laptop-3	192.168.20.102

Observation

Each device automatically received a unique IP address from the configured DHCP pool.

Result

The DHCP server successfully assigned valid IP addresses to all client devices.

3. Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

Aim

To configure DNS
custom hostname.

name resolution for a

DNS Configuration

Hostname	IP Address
musicplayer.local	192.168.20.10

Verification

Typing the following address into the web browser:

http://musicplayer.local opened the local web page
hosted at 192.168.20.10.

Observation

The custom hostname resolved successfully to the local server IP address.

Result

The DNS service correctly translated the hostname into its corresponding IP address.

4. Create two VLANs (e.g., VLAN 10: "Workstations", VLAN 20: "Smart Devices") on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.

Hint: Use ping or file sharing to test connectivity between devices in different VLANs.

Aim

To create VLANs and verify communication within and between VLANs.

VLAN Configuration

VLAN ID	VLAN Name	Devices
VLAN 10	Workstations	PC-1, PC-2
VLAN 20	Smart Devices	PC-3, PC-4

Connectivity Test

Source	Destination	Result
PC-1	PC-2	Successful
PC-3	PC-4	Successful

PC-1	PC-3	Failed
PC-2	PC-4	Failed

Observation

Devices within the same VLAN communicated successfully, while communication between different VLANs was blocked because inter-VLAN routing was not configured.

Result

The VLAN configuration successfully isolated the two networks.

5. Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

Aim

To configure NAT on a router and provide Internet access to LAN devices. NAT Configuration

Interface	Configuration
LAN Interface	192.168.10.1/24
WAN Interface	203.0.113.2
NAT	Enabled (PAT/Overload)

Verification

Command Used:

ping www.wikipedia.org

Observation

The ping command successfully reached the external server, indicating that NAT translated the private IP addresses into the public IP address.

Result

The router successfully provided Internet access to all LAN devices using Network Address Translation (NAT).